

Cryptography Assignment - 4

Decrypt the following ciphertext using the hill cipher.

1. Cipher text : HLDI QIHJ DXWQ CMAG

$$\text{Key} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

Use ($A=0, B=1, \dots, Z=25$)

Find the plain text:

Ans: (i) Find the determinant of K .

$$\det(K) = 3(7) - 5(2) = 21 - 10 = 11$$

(ii) Find K^{-1}

$$K^{-1} = (\det K)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{26}$$

$$11^{-1} \pmod{26} : 26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$26 + 11(-2) + [11 + 4(-2)](-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + [26 + 11(-2)](2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

$$11^{-1} \pmod{26} = 19 \pmod{26} = 19$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

(iii) Divide the ciphertext into pairs

HL DI QI HJ DX WQ CM AG

Formula: $P = K^{-1} C \pmod{26}$

$$HL = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \pmod{26} = \begin{bmatrix} 175 \\ 118 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 14 \end{bmatrix} = \begin{matrix} T \\ O \end{matrix}$$

$$DT = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \pmod{26} = \begin{bmatrix} 275 \\ 122 \end{bmatrix} \pmod{26} = \begin{bmatrix} 15 \\ 18 \end{bmatrix} = \begin{matrix} P \\ S \end{matrix}$$

$$QI = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \pmod{26} = \begin{bmatrix} 160 \\ 184 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{matrix} L \\ C \end{matrix}$$

$$HJ = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} \pmod{26} = \begin{bmatrix} 147 \\ 108 \end{bmatrix} \pmod{26} = \begin{bmatrix} 17 \\ 4 \end{bmatrix} = \begin{matrix} R \\ E \end{matrix}$$

$$OX = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} \pmod{26} = \begin{bmatrix} 331 \\ 142 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 12 \end{bmatrix} = \begin{matrix} T \\ M \end{matrix}$$

$$WQ = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} \pmod{26} = \begin{bmatrix} 290 \\ 278 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 18 \end{bmatrix} = \begin{matrix} L \\ S \end{matrix}$$

$$CM = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} \pmod{26} = \begin{bmatrix} 174 \\ 78 \end{bmatrix} \pmod{26} = \begin{bmatrix} 18 \\ 0 \end{bmatrix} = \begin{matrix} S \\ A \end{matrix}$$

$$AG = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} \pmod{26} = \begin{bmatrix} 84 \\ 30 \end{bmatrix} \pmod{26} = \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{matrix} G \\ K \end{matrix}$$

$\therefore$ Plaintext = TOP SECRET MESSAGE

2. Consider the Hill Cipher key matrix

$$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$$

Ciphertext : RXAODY

Find the plaintext for the above cipher text

Ans: (i) Find determinant of K.

$$\det(K) = 24(14) - 5(19) = 336 - 95 = 241$$

$$241 \pmod{26} = 7$$

(ii) Find K^{-1}

$$K^{-1} = (\det K)^{-1}$$

$$K^{-1} = (\det K)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{26}$$

$$(\det K)^{-1} \} = 7^{-1} : 26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$[26 + 7(-3)] + [7 + 5(-1)](-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-3) + 7(-2) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$7^{-1} \bmod 26 = 15 \bmod 26 = 15$$

$$K^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} (\bmod 26)$$

$$K^{-1} = \begin{bmatrix} 210 & -285 \\ -75 & 360 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

(iii) Divide cipher text into pairs:

$$RX \quad AO \quad DY \quad \text{Formula: } P = K^{-1}C \bmod 26$$

$$RX = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} \bmod 26 = \begin{bmatrix} 57 \\ 557 \end{bmatrix} \bmod 26 = \begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{matrix} F \\ L \end{matrix}$$

$$AO = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} \bmod 26 = \begin{bmatrix} 14 \\ 308 \end{bmatrix} \bmod 26 = \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{matrix} O \\ W \end{matrix}$$

$$DY = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} \bmod 26 = \begin{bmatrix} 30 \\ 537 \end{bmatrix} \bmod 26 = \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \begin{matrix} E \\ R \end{matrix}$$

FLOWER

$\therefore$ The plaintext = FLOWER.

3. Analyze the following classical ciphers with respect to confusion and diffusion. Complete the table and justify your classification based on how each cipher transforms the plaintext.

Ans:	Cipher	Changes character/ symbol value?	Changes character position?	Confusion/ Diffusion	Justification
	Shift cipher	Yes	No	Confusion only	Substitutes each letter via a fixed shift, position unchanged. Monoalphabetic so frequency patterns are only related.
	Multiplicat- ive cipher	Yes	No	Confusion only	Substitutes letters via a fixed multiplier, position unchanged. Still monoalpha- batic still frequency vulnerable.
	Affine cipher	Yes	No	Confusion only	Combines shift + multiplication into one substitution mapping. Still monoalphabetic only positions are untouched.
	Transposition cipher	No	Yes	Diffusion only	Rearranges character position, symbol values untouched. Letter frequencies survive, but positional patterns are scattered.

4. Based on your observations, explain the relationship between substitution and confusion and between transposition and diffusion.

Ans: Substitution $\leftrightarrow$ Confusion: Substitution ciphers replace symbol values, obscuring the key eg ciphertext relationship. This is confusion. Shift, multiplicative and affine ciphers are all substitutions are confusion only, no diffusion.

Transposition $\leftrightarrow$ Diffusion: Transposition ciphers reorder symbol positions without changing values, spreading plaintext structure across the ciphertext. This is diffusion $\neq$ no confusion, since values stay intact.

Transposition Encryption:

Plaintext: THE ARROW IS POINTING NORTH DIRECTION
(32 Letters)

(a) Simple transposition, key = 4

T H E A	Ciphertext:
R R O W	TRIINRIT HRSNGTRI EOPTNHED
I S P O	AWOIODCN
I N T I	
N G N O	
R T H D	
I R E C	
T I O N	

(b) Keyword Transposition, Keyword = ATLAS

alphabetically, A(1), A(4), L(3), S(5), T(2)

Grid (5 columns, padded with x to fill row 7):

T H E A R	Ciphertext:
R O W I S	TRPIRRO AINNDTX EWIGHCX
P O I N T	RSTOIIX HOO NTEN.
I N G N O	
R T H D I	
R E C T I	
O N X X X	

(c) Double Transposition, Key 1 = BREAK, Key 2 = LOG.

Pass 1 (BREAK): A(4), B(1), E(3), K(5), R(2)

A

Intermediate ciphertext:

A I N N D T X T R P I R R O E W I G H C X R S T O I I X H O O N T E N
 ⇒ A I N N D T X T R P I R R O E W I G H C X R S T O I I X H O O N T E N

Pass 2 (LOG): G(3), L(1), O(2)

A I N

N D T

X T R

P I R

R O E

W I G

H C X

R S T

O I I

X H O

O N T

E N X

Cipher text:

N T R R E G I X T I O T X A N X P R W H R O X D E

I D T I D I C S I H N N

Transposition Decryption:

(a) Simple, key = 10

Ciphertext (60 letters) is split into chunks of 6 rows and reassembled into columns 1 → 10.

T R O O P S H E A D
 I N G W E S T N E E
 D M O R E S U P P L
 I E S S E N D G E N
 E R A L D U B O I S
 M E N T O A I D X X

Plaintext:

TROOPS HEADING WEST NEED MORE SUPPLIES SEND
 GENERAL DUBOIS MEN TO AID.

(b) Keyword = SECRET

T H I S I S
 A P L A I N
 T E X T F O
 R D E M O N
 S T R A T I

G(3), E(2), E(5), R(4), S(1), T(6)

1 0 N X X X

Plaintext:

THIS IS A PLAIN TEXT FOR DEMONSTRATION.

cc) keyword = PLAIN. Cipher: V E Y E A X

PLAIN $\rightarrow$ A(3), I(4), L(2), N(5), P(1)

H A V E A

G R E A T

D A Y X X

Plaintext: HAVE A GREAT DAY.

5. Find the Inverse permutation for.

(i) 4 3 1 5 2.

$P = 4 \ 3 \ 1 \ 5 \ 2.$

Position :	1	2	3	4	5
P :	4	3	1	5	2

$P^{-1} = \underline{\underline{(3, 5, 2, 1, 4)}}$

(ii) $P = (7, 4, 1, 3, 5, 2, 6)$

Position :	1	2	3	4	5	6	7
P :	7	4	1	3	5	2	6

$P^{-1} = \underline{\underline{(3, 6, 4, 2, 5, 7, 1)}}$

(iii) $P = (4, 1, 3, 2)$

Position :	1	2	3	4
P :	4	1	3	2.

$P^{-1} = \underline{\underline{(2, 4, 3, 1)}}$

(iv) $P = (3, 5, 1, 4, 2)$

Position :	1	2	3	4	5
P :	3	5	1	4	2

$P^{-1} = \underline{\underline{(3, 5, 1, 4, 2)}}$

7.